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CC 207B Unit 5 Assignment

Name: James C. Remegio Date: 10/14/2026

Year and Section: BSIT 3A

Note: Upload your Flutter project to your assigned folder here -> https://github.com/WVSU-BSIT-Batch-2028/CC-207B_26-27_Activities. Read the instructions on submitting, including the file name/folder conventions in the GitHub repository.

I. CICT Org Fair App Development. The CICT is holding an Org Fair. Students browse the student organizations on their phone, open one to read about it, and sign up with a short form. Most of this app is already written for you. Your job is to **finish 14 small pieces**, **explain how the rest works**, and **run experiments** that show what each Flutter idea from 5.1 and 5.2 really does. You will write about 40 lines of code. **(35 points)**

Get the initial Flutter project source code from: https://github.com/WVSU-BSIT-Batch-2028/CC-207B_26-27_Samples/tree/main/Unit%205/assignment_org_fair

Here is the table specifying the code changes you need to perform:

File	What it does	What you do
`lib/main.dart`	Starts the app and sets the theme	Read (E1)
`lib/data/orgs.dart`	The `Org` class and a pretend web service, `OrgRepository`, that waits about 2 seconds and returns orgs, nothing, or an error	Read only. Do not change
`lib/models/registration.dart`	The `Committee` enum and the `Registration` class the form returns	Read
`lib/widgets/state_message.dart`	One layout for the loading, error, empty and no-results messages	Read
`lib/widgets/org_card.dart`	One org in the list	Read (E6)
`lib/widgets/org_banner.dart`	The banner photo, with loading and error handling	Read (E8)
`lib/screens/org_list_screen.dart`	Loads the orgs, search, list or grid, swipe and long-press	TODO 1 to 7; E2 to E5, E7

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`lib/screens/org_detail_screen.dart`	One org with its banner and a Join button	TODO 8; E9
`lib/screens/join_form_screen.dart`	The sign-up form with 8 kinds of input	TODO 9 to 14; E10, E11
`test/org_fair_test.dart`	7 automatic tests. Each one names the TODOs it checks	Run to check your work

Note: Run the command “flutter test” at any time. All tests should pass before submitting the assignment.

Each TODO is a comment in the code with a placeholder line under it. Replace the placeholder with your code. Most need 1 to 6 lines. In `org_list_screen.dart` the loading state is already written; use it as a pattern for TODO 4 and 5.

#	File	What to do	Slide topic
1	org_list_screen	Start loading the orgs once in `initState()`	StatefulWidget Lifecycle
2	org_list_screen	`_reload()`: fetch again inside `setState()`	What Does setState() Do?
3	org_list_screen	`_toggleFavorite()`: add or remove the id inside `setState()`	setState; 5.1 Set
4	org_list_screen	Error state with a **Try again** button	The Four States of Content
5	org_list_screen	Empty state with a **Refresh** button	The Four States of Content
6	org_list_screen	Search box `onChanged` saves the text with `setState()`	Getting Values from Users
7	org_list_screen	Wrap each card in a `Dismissible`	Dismissibles
8	org_detail_screen	Join button: disabled when already a member	Affordance and Feedback
9	join_form_screen	`validate()` then `save()` the form	Handling forms with validation
10	join_form_screen	Send the registration back with `Navigator.pop`	Navigation widgets
11	join_form_screen	Email validator with helpful messages	When Should Errors Appear?
12	join_form_screen	Student ID: digits-only formatter and validator	Input Validations (formatters)
13	join_form_screen	Slider, Switch and Checkbox `onChanged` with `setState()`	Inputs Follow One Loop
14	join_form_screen	Submit stays disabled until the checkbox is ticked	Affordance and Feedback

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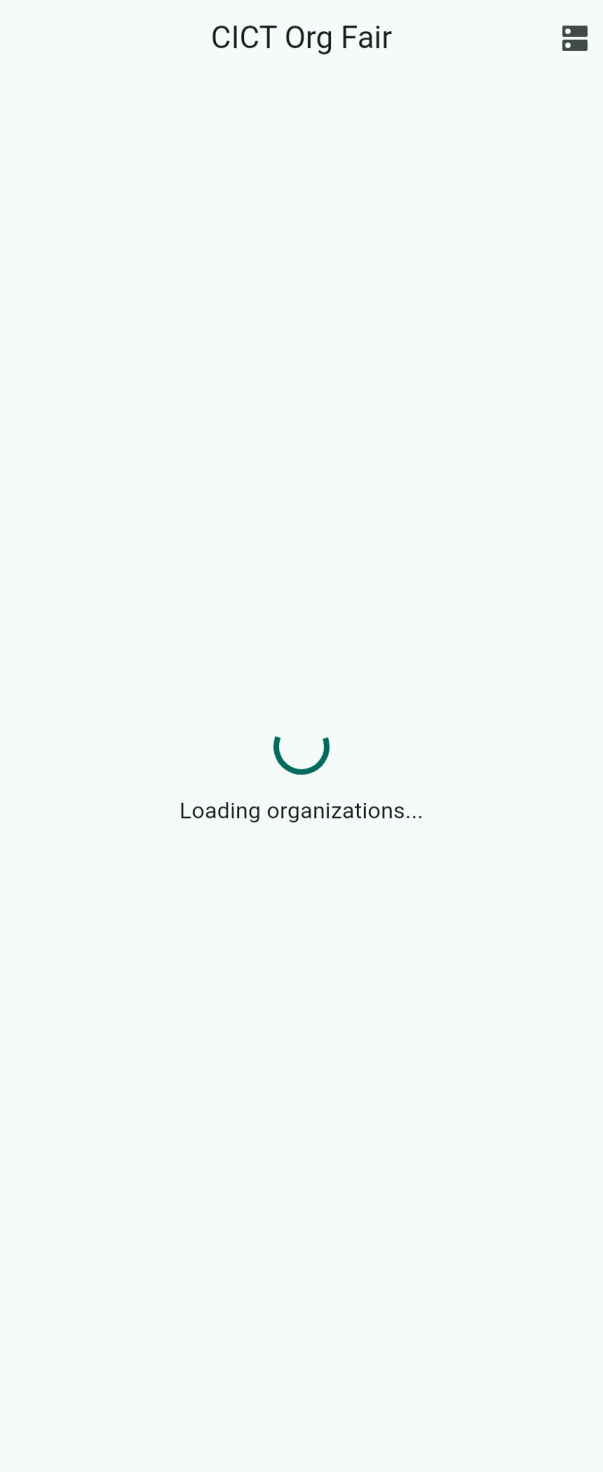
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Evidence (Provide screenshots of the application to show that it is working):

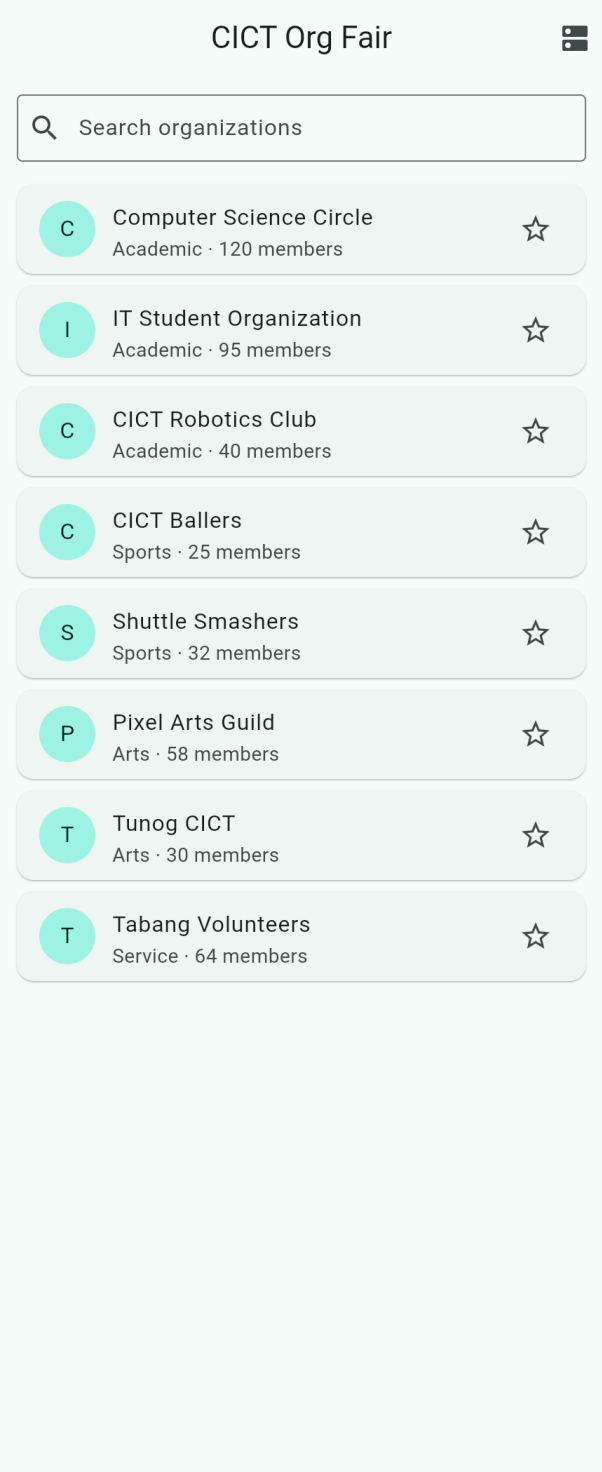
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01. Loading organizations



The loading indicator appears while a repository request is pending. TODOs 1-2.

02. Organization list



A successful request displays all eight organizations in the phone-width layout. TODO 1.

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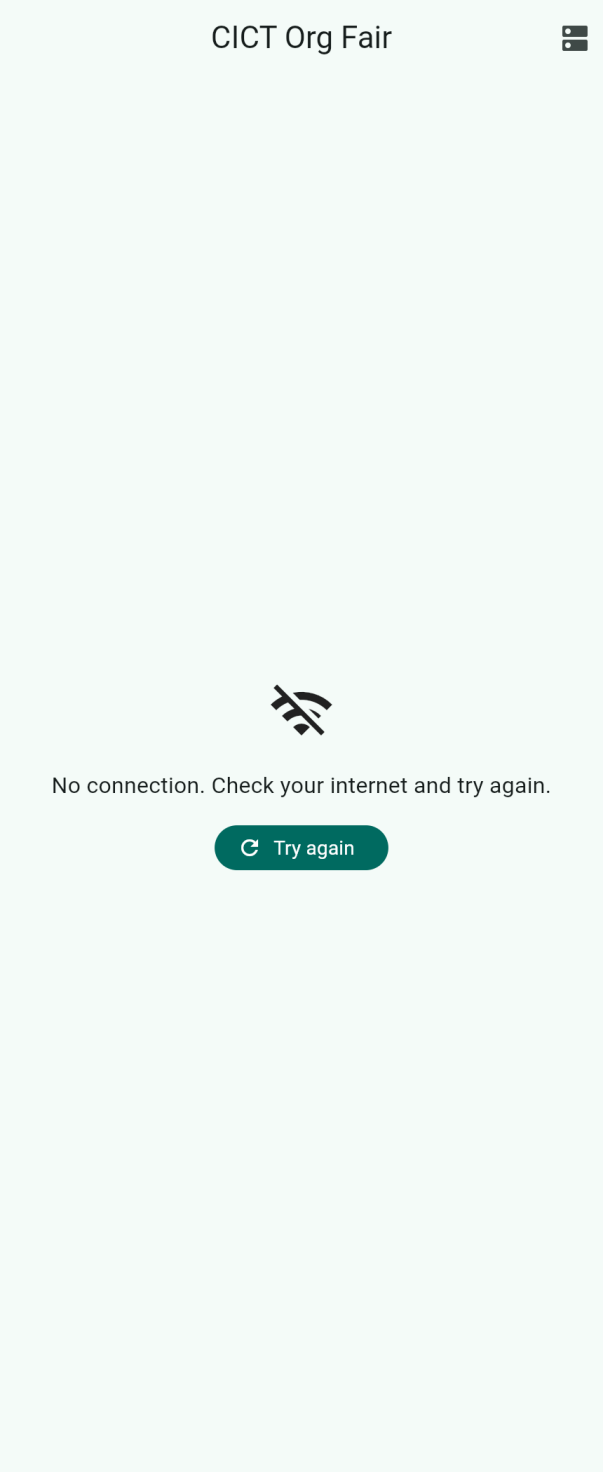
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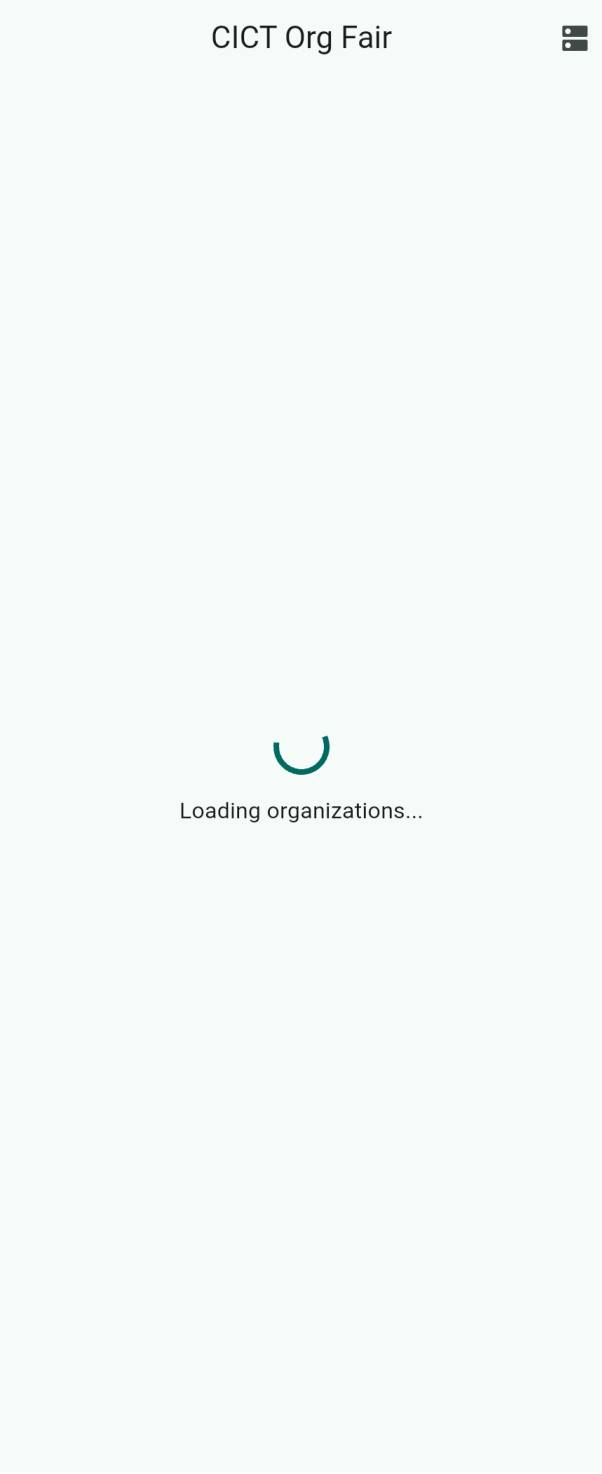
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03. Error feedback



Server: error displays a helpful connection message and the Try again action. TODO 4.

04. Retry starts another request



Pressing Try again returns the screen to loading. With error mode unchanged, it returns to the error state. TODO 2; E3.

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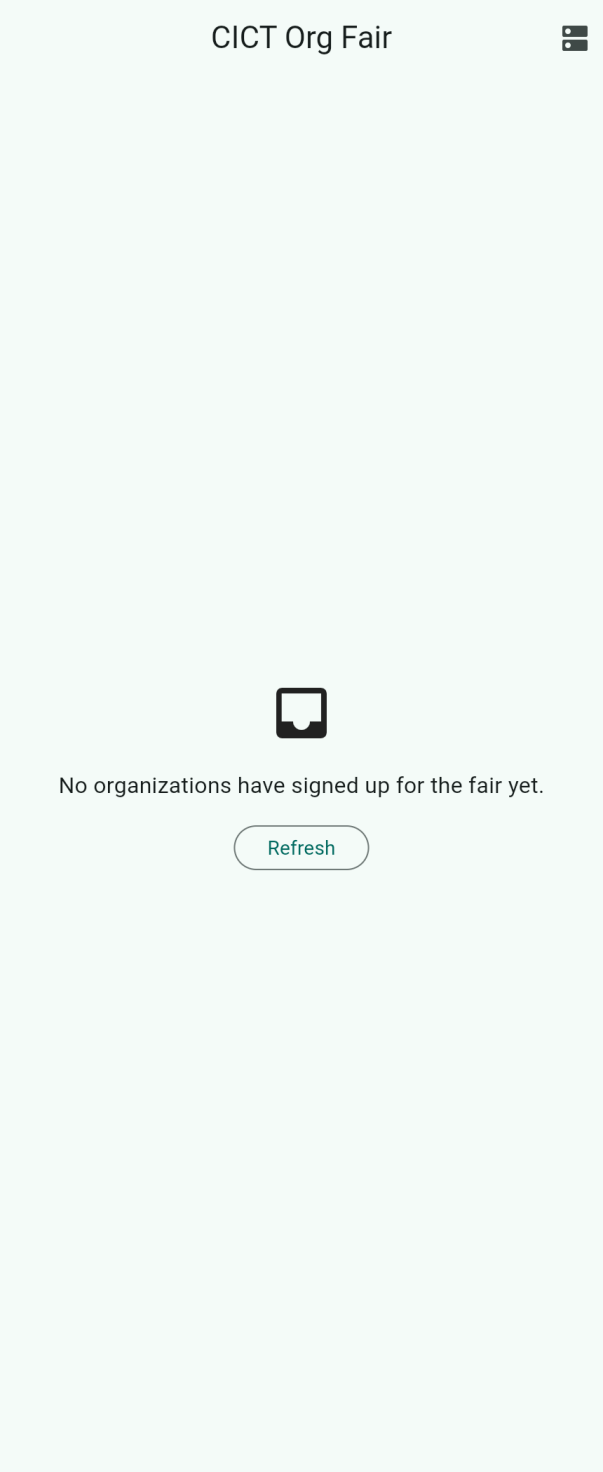
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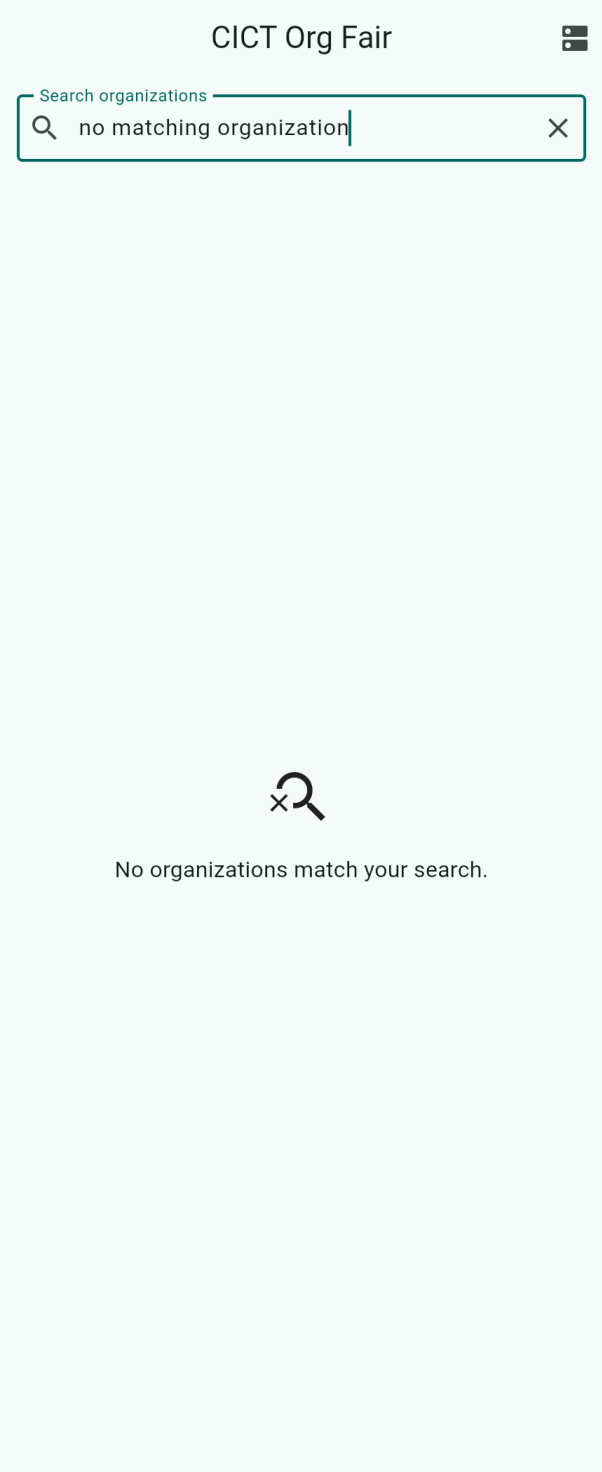
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05. Empty repository result



Server: empty displays the empty-state message and Refresh action. TODO 5.

07. Search with no matches



An unmatched query displays a distinct no-results message. TODO 6.

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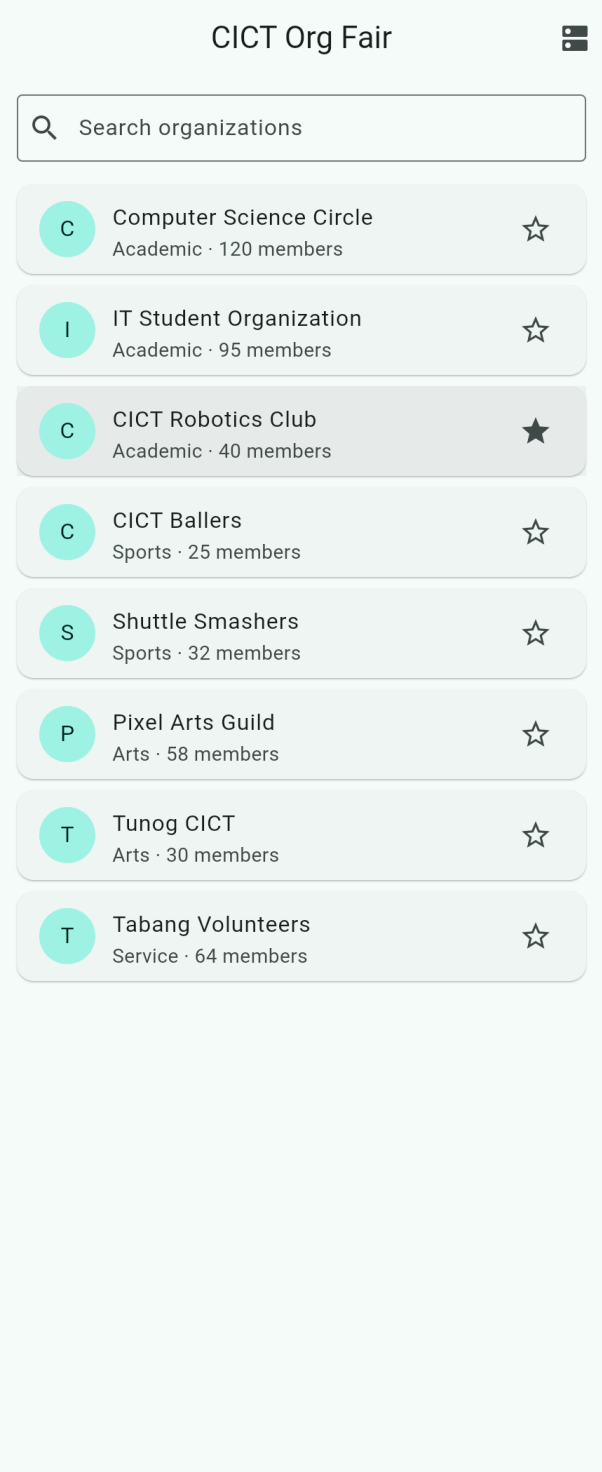
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06. Search filters the list



Typing robotics leaves CICT Robotics Club as the only matching organization. TODO 6; E4.

08. Long-press favorite



Long-pressing CICT Robotics Club fills its star. The parent screen stores the favorite state. TODO 3; E6.

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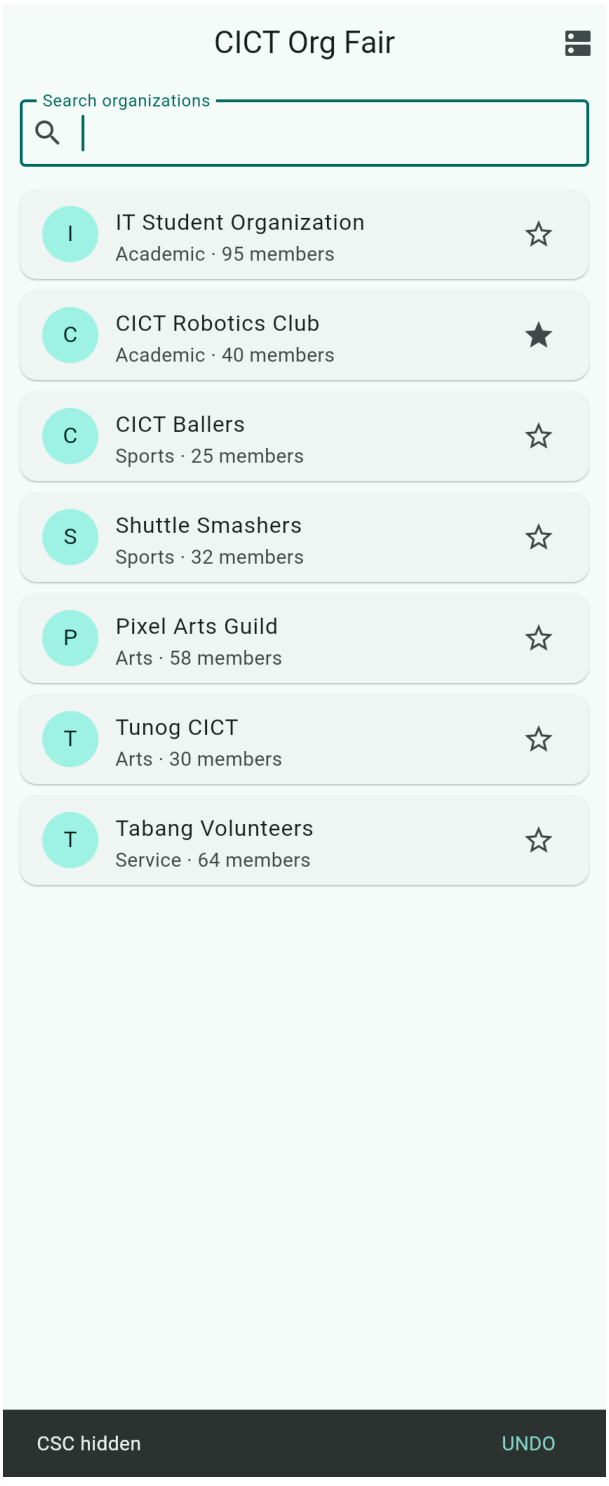
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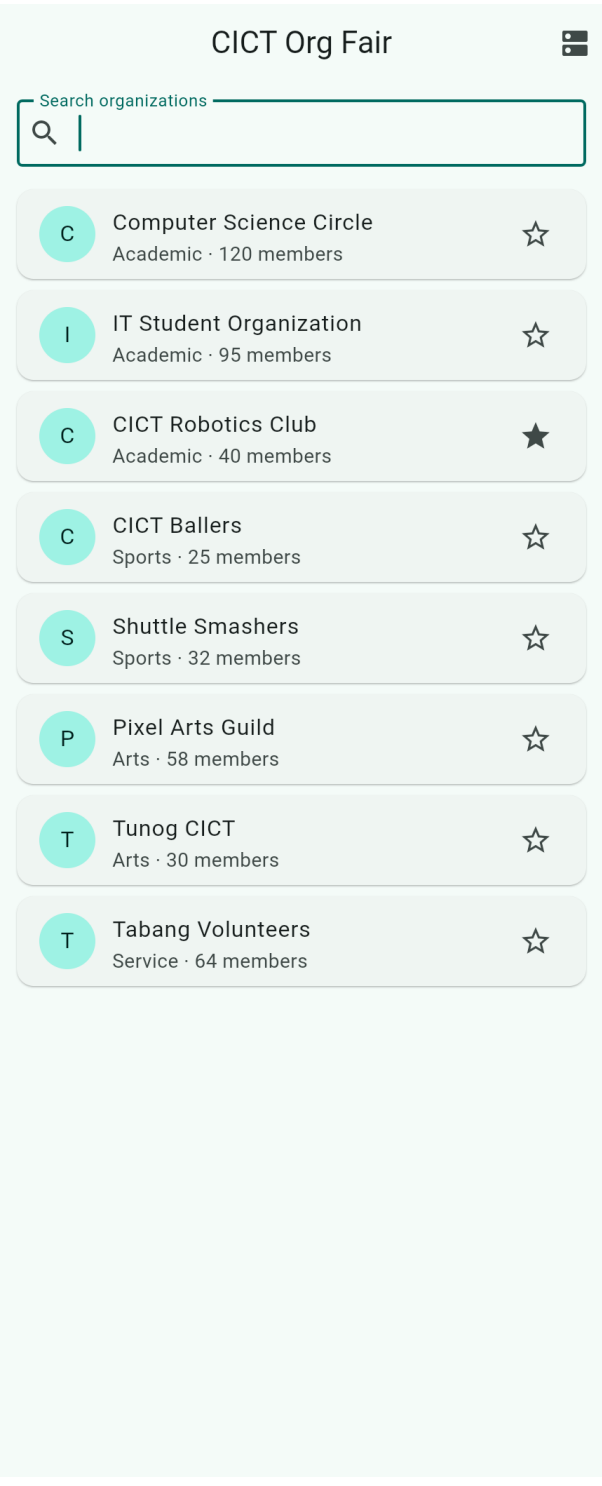
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09. Swipe to hide



Swiping Computer Science Circle removes its card and displays CSC hidden with UNDO. TODO 7.

10. Undo restores the card



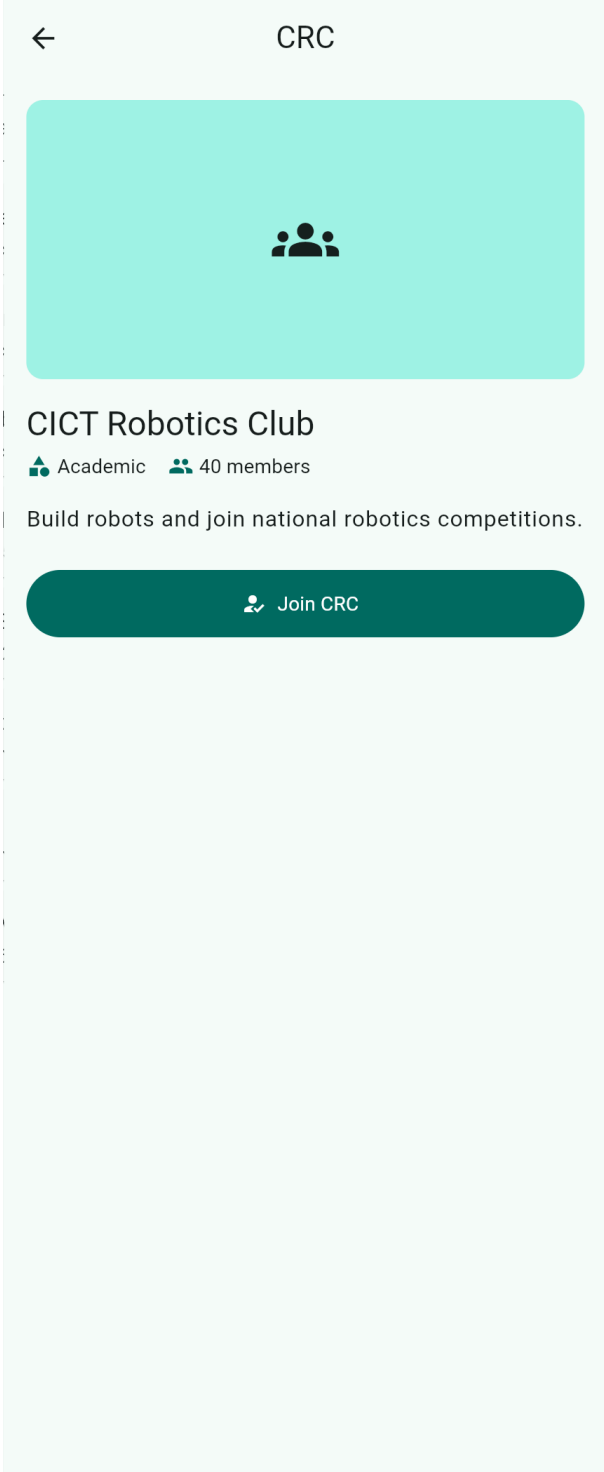
Pressing UNDO restores Computer Science Circle; the Robotics favorite remains selected. TODO 7.

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Evidence (Provide screenshots of the application to show that it is working):

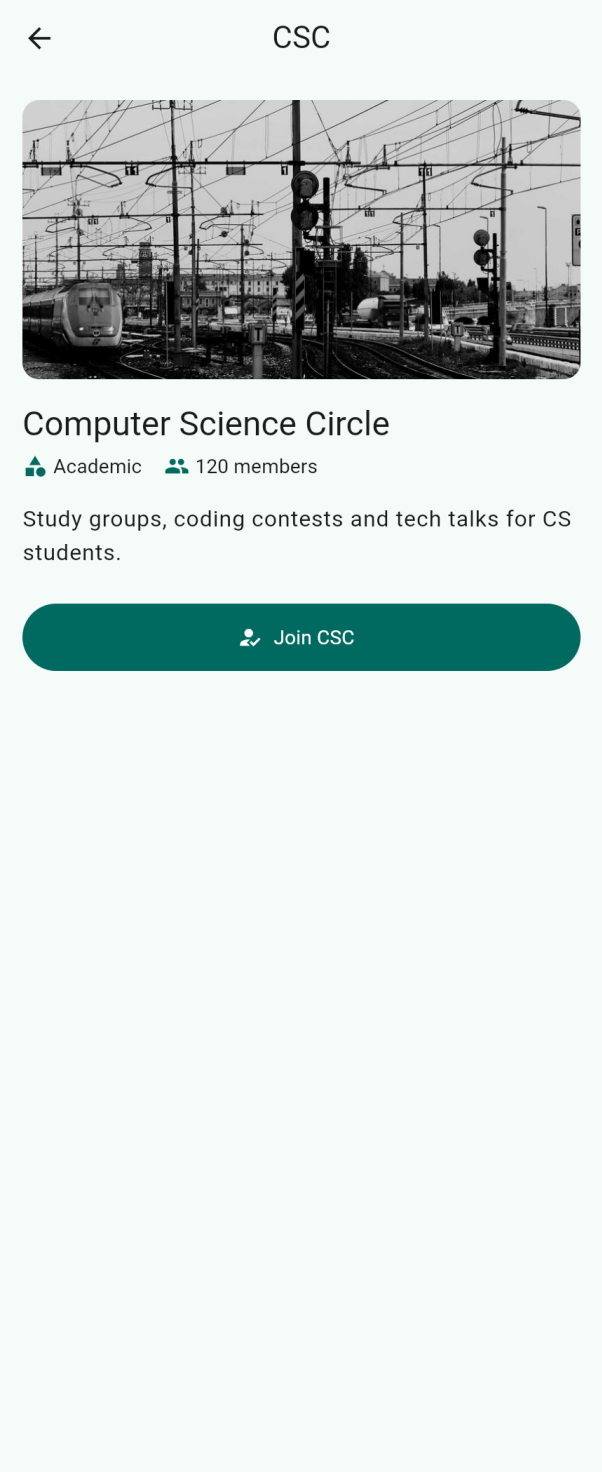
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11. Details and missing banner



Opening Robotics shows its description, member count, Join button, and placeholder for its null banner URL. E8.

19. Network banner loaded



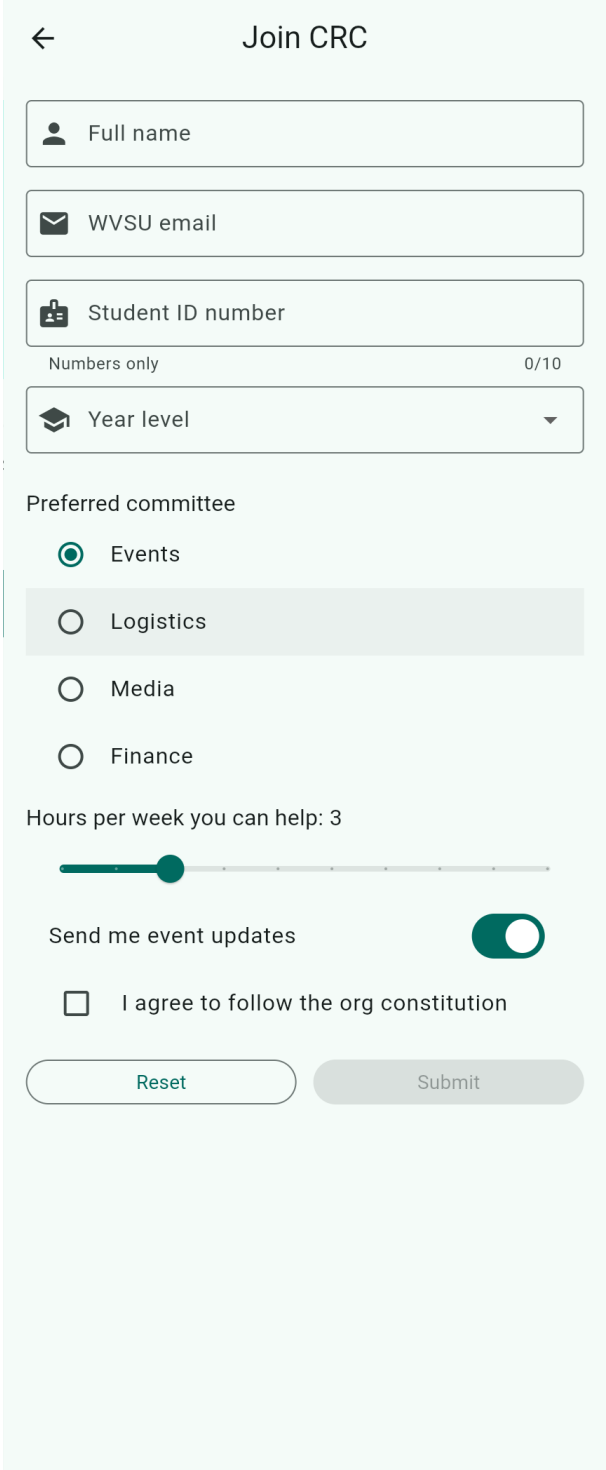
Computer Science Circle displays its downloaded banner image and organization details. E8.

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Evidence (Provide screenshots of the application to show that it is working):

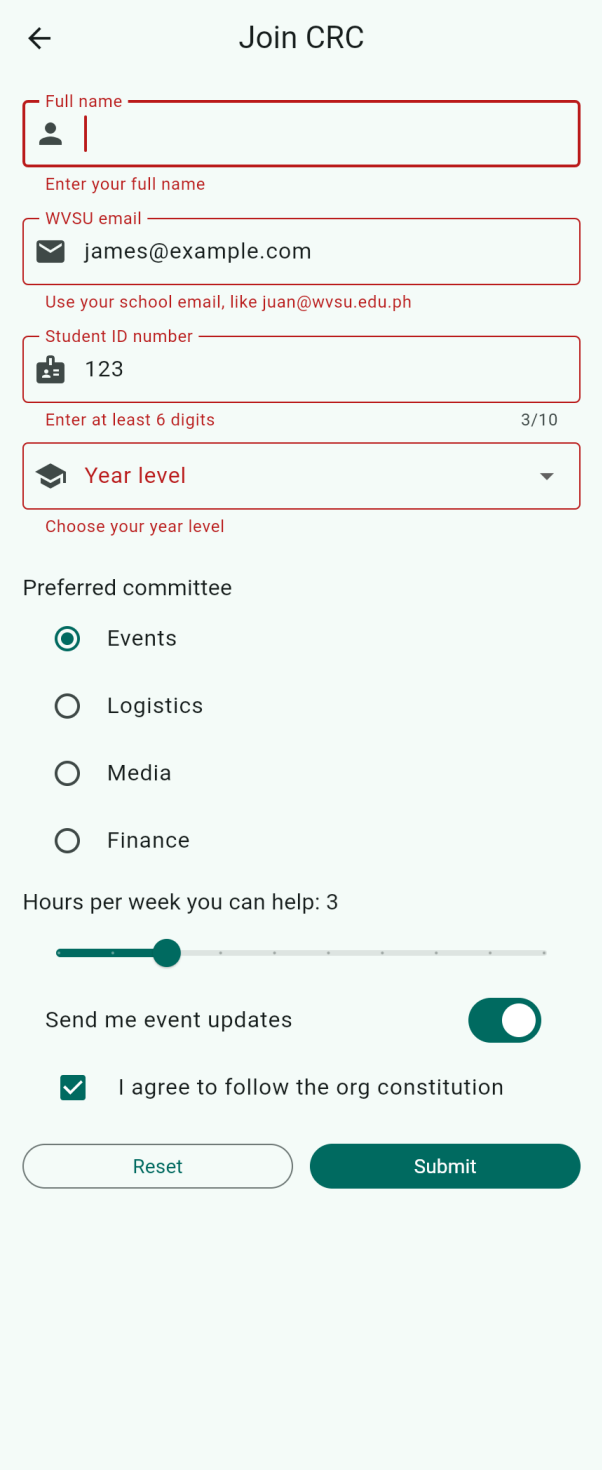
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12. Initial form state



The untouched form has no validation errors. Submit is disabled while the agreement is unchecked. TODO 14; E11.

13. Validation and digits only



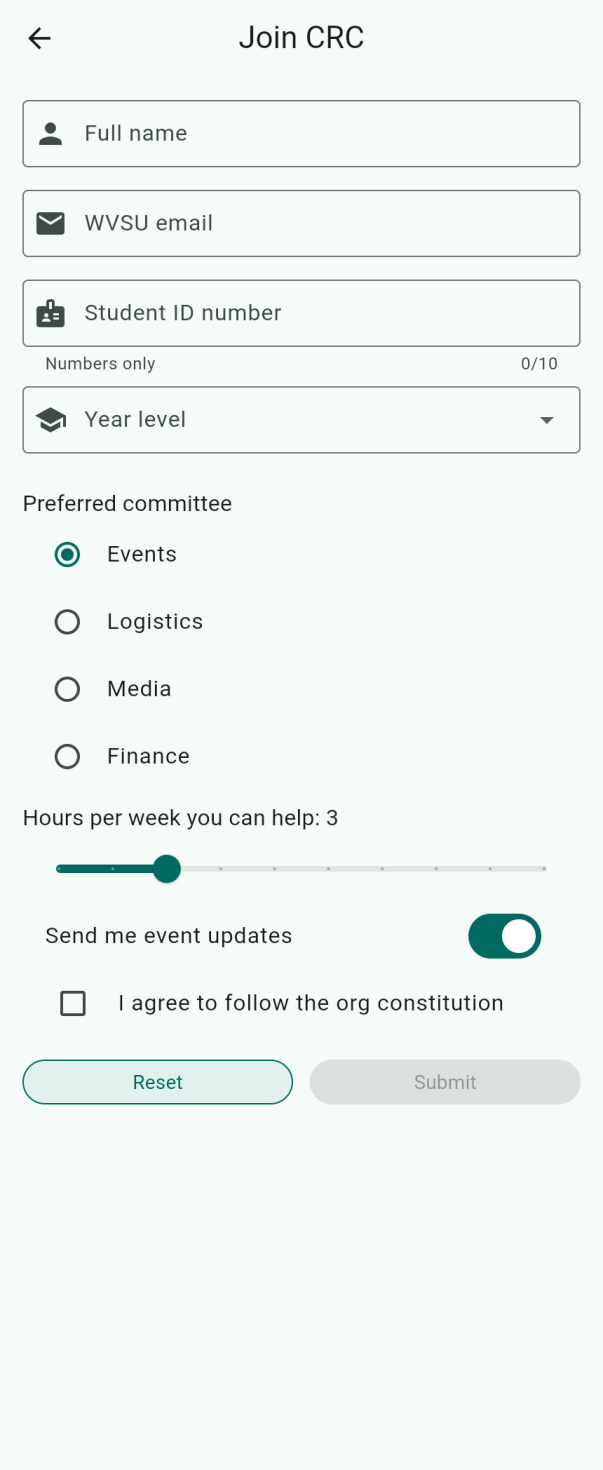
Invalid submission shows name, school-email, ID-length and year-level errors. Typing 12ab3 leaves only 123. TODOs 9, 11-12.

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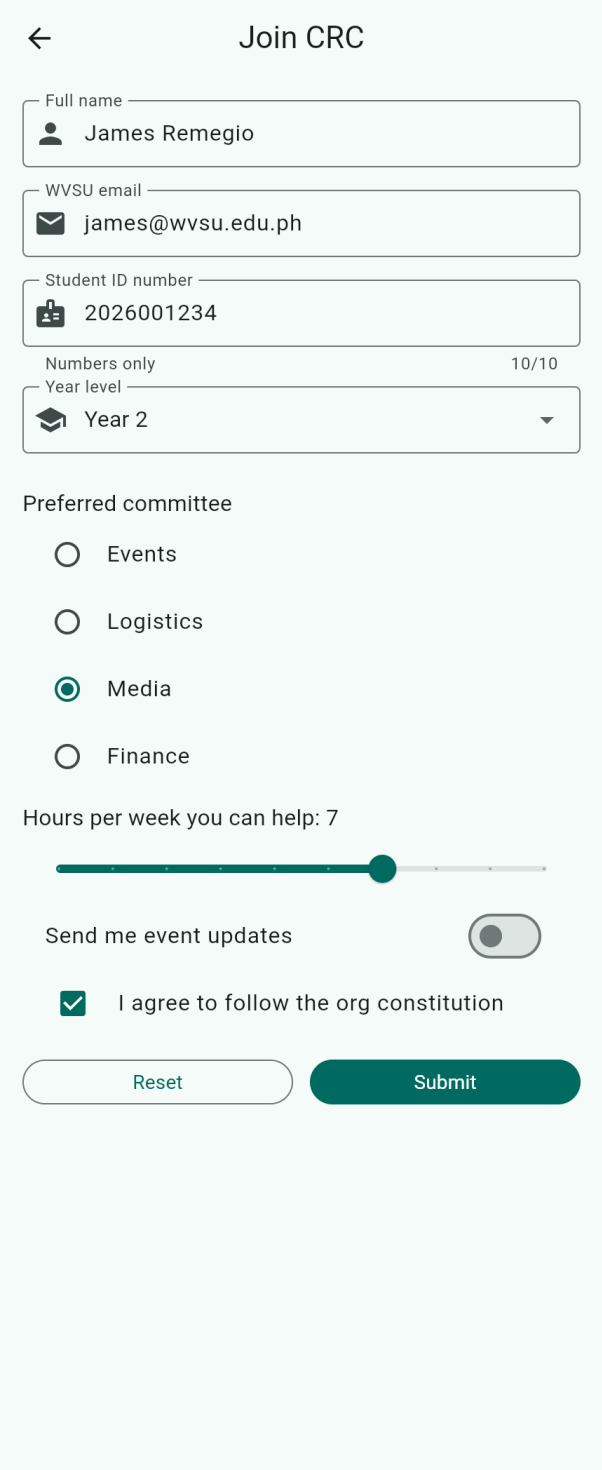
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14. Reset clears the form



Reset clears values and validation errors and restores Events, 3 hours, updates on, and an unchecked agreement.

15. Valid form and changed inputs



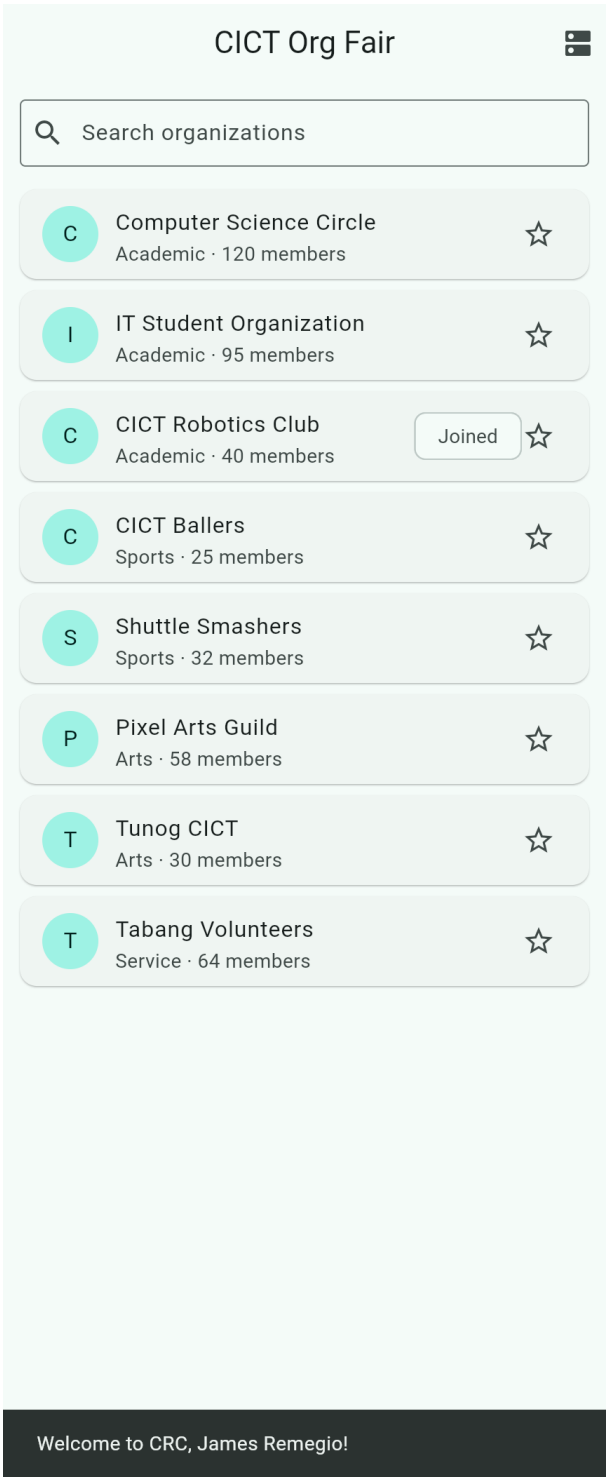
Demo data is entered; Year 2, Media, 7 hours and updates off are selected. Checking the agreement enables Submit. TODOs 11-14.

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Evidence (Provide screenshots of the application to show that it is working):

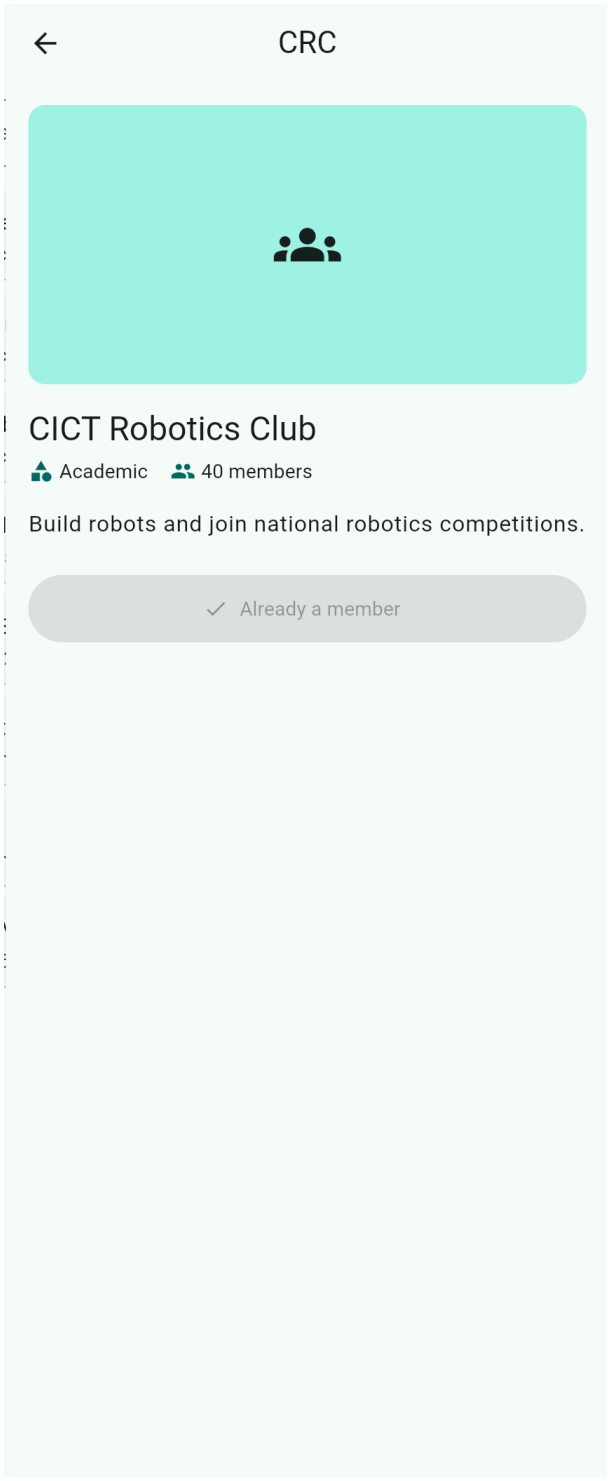
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16. Registration returned to list



Submitting valid data returns to the list, displays Welcome to CRC, James Remegio! and marks Robotics as Joined. TODOs 9-10; E7, E9.

17. Duplicate joining disabled



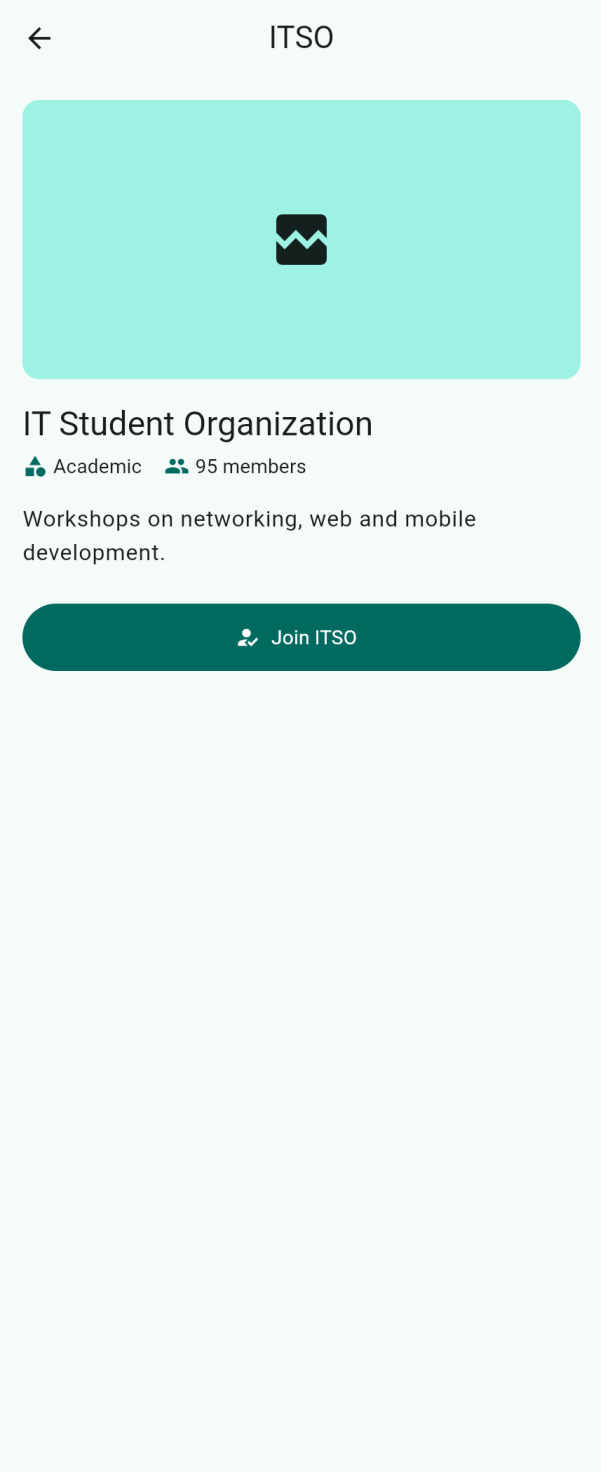
Reopening the joined organization displays a disabled Already a member button. TODO 8.

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Evidence (Provide screenshots of the application to show that it is working):

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20. Failed-image fallback



The ITSO image request was deliberately blocked in the browser. The app displays its broken-image placeholder and remains usable. E8.

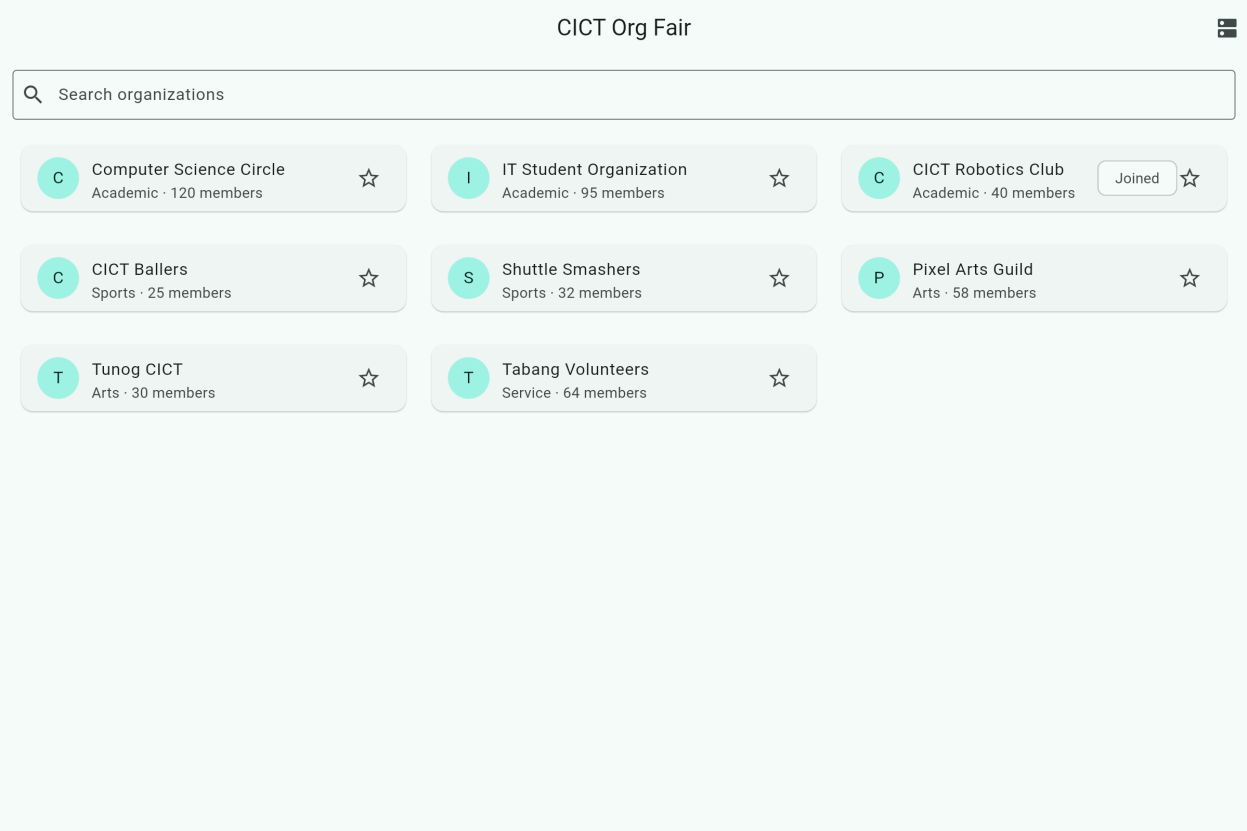
The ITSO banner request was deliberately blocked in the browser to test the errorBuilder. The application displayed its broken-image placeholder and remained usable. No application code or repository data was changed for this capture.

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Evidence (Provide screenshots of the application to show that it is working):

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18. Responsive grid layout



At a 1200-pixel viewport width, the same screen displays three columns. The joined state is retained. E5. Captured in Chrome at 1200 x 800; portrait views use 430 x 1050. Registration values are sample test data.

Verification: all 7 Flutter tests passed; flutter analyze found no issues.

Additional code experiments (performed on a temporary copy):

E5: Removing Expanded produced "Vertical viewport was given unbounded height." Restoring Expanded resolved the layout error.

E11: onUserInteraction showed no initial errors and validated after editing. Switching to always showed required-field errors immediately.

The submitted source retains Expanded and onUserInteraction.

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II. CICT Org Fair Code Analysis. The code has 11 comments that start with `EXPLAIN E1` to `EXPLAIN E11`. Put your answers in the available column for each row. (2 points each)

#	Where	The question is about	Answer
E1	main.dart	Why OrgFairApp is stateless but OrgListScreen is stateful?	OrgFairApp has no changing local state. OrgListScreen owns the future, query, favorites and registrations, which rebuild through setState().
E2	org_list_screen: dispose()	Why controllers must be disposed?	Dispose releases controller listeners/resources, preventing leaks when the screen is removed.
E3	org_list_screen: FutureBuilder	The order of states when the user presses Try again?	Error -> loading -> error again. Try again replaces the future; error mode keeps failing until the mode changes.
E4	org_list_screen: _buildOrgs	How the list updates while the user types?	onChanged saves _query in setState(); rebuilding filters the list with the new query.
E5	org_list_screen: Expanded	Constraints go down, sizes go up?	Expanded bounds the viewport height; without it, height is unbounded. Constraints go down; sizes go up.
E6	org_card.dart	Where the favorite star's state lives?	The parent owns _favoriteIds; setState rebuilds OrgCard with the updated isFavorite value.
E7	org_list_screen: _openDetail	How data comes back from another screen?	pop(registration) completes the push() future; the detail passes it to the list to save.
E8	org_banner.dart	Null safety and the loading and error builders?	A null URL shows a placeholder. loadingBuilder shows progress, then the image; errorBuilder shows a broken-image icon on failure.
E9	org_detail_screen: _join	Why check context.mounted after await?	The screen may be removed during await; mounted guards using an invalid context.
E10	join_form_screen: _formKey	What a GlobalKey<FormState> gives you?	GlobalKey<FormState> accesses this FormState: validate() checks fields, save() runs onSaved, and reset() restores initial values.
E11	join_form_screen: autovalidateMode	When errors should appear?	onUserInteraction validates after input or on submit, avoiding initial errors. always validates even untouched fields from the start.

Determine what type of widgets and whether they are either stateless or stateful. (2 points each)

Feature	Widget(s) used	Stateless or Stateful? Why?
Org list or grid	OrgListScreen; ListView.builder / GridView.builder	Stateful screen: owns future, search and sets; setState rebuilds the list/grid.
Favorite star	OrgCard; Icon	Stateless card/icon: favorite state belongs to the parent and is passed in.
Banner image	OrgBanner; Image.network	OrgBanner is stateless; Image is stateful and handles asynchronous loading.
Submit button	FilledButton; JoinFormScreen	FilledButton and form are stateful; form agreement controls whether Submit is enabled.